

vaadin}>

Building **AI-Native** Enterprise UIs with Vaadin & Spring

Let Users Talk to Their Data



Sebastian Kuehnau
Vaadin Enthusiast

Disclaimer

Who is talking?

Sebastian Kuehnau

+20y in web- & java development

+10y at Vaadin

several technical roles

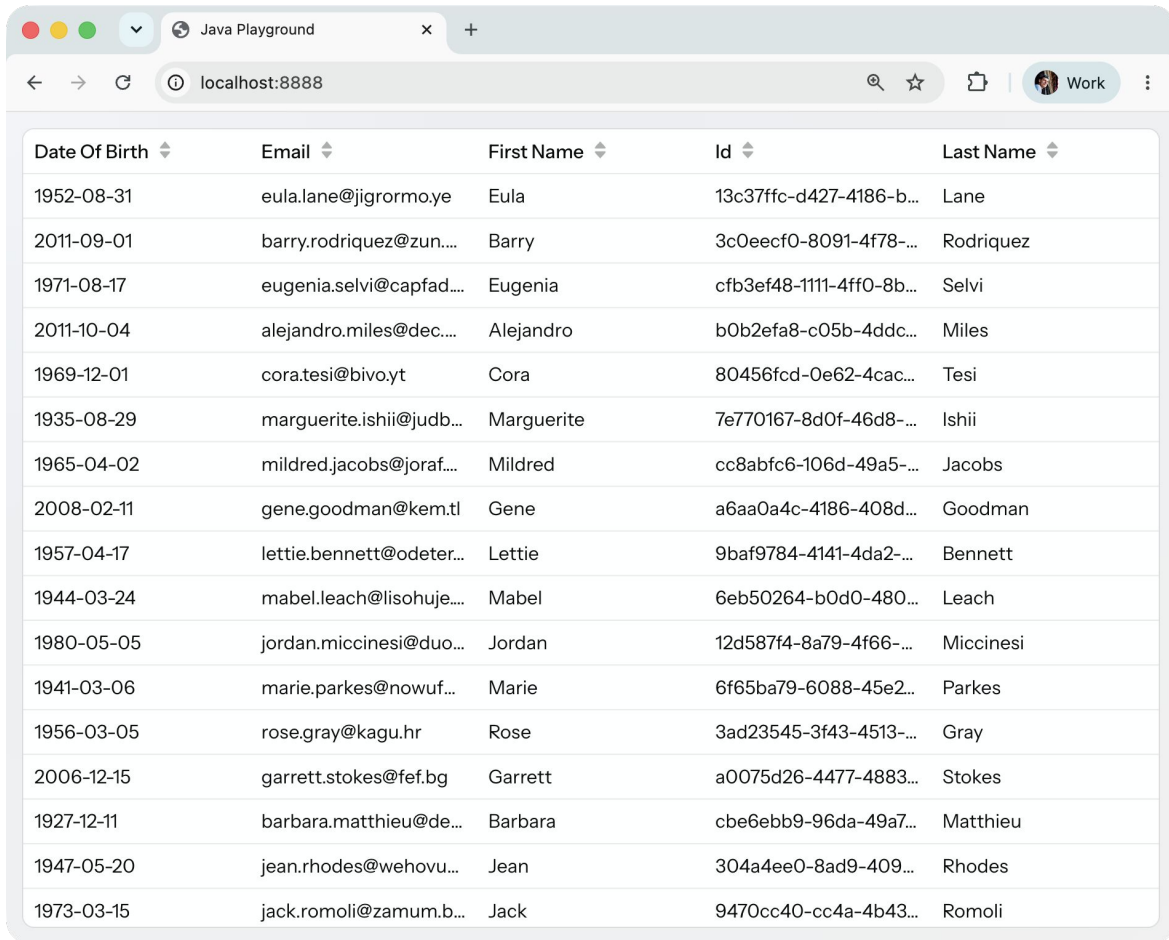
now developer advocate

Off-work mostly outdoors;



Focus on data

Simple Grid



Date Of Birth	Email	First Name	Id	Last Name
1952-08-31	eula.lane@jigrormo.ye	Eula	13c37ffc-d427-4186-b...	Lane
2011-09-01	barry.rodriquez@zun....	Barry	3c0eecf0-8091-4f78-...	Rodriquez
1971-08-17	eugenia.selvi@capfad....	Eugenia	cfb3ef48-1111-4ff0-8b...	Selvi
2011-10-04	alejandro.miles@dec....	Alejandro	b0b2efa8-c05b-4ddc...	Miles
1969-12-01	cora.tesi@bivo.yt	Cora	80456fcd-0e62-4cac...	Tesi
1935-08-29	marguerite.ishii@judb...	Marguerite	7e770167-8d0f-46d8-...	Ishii
1965-04-02	mildred.jacobs@joraf....	Mildred	cc8abfc6-106d-49a5-...	Jacobs
2008-02-11	gene.goodman@kem.tl	Gene	a6aa0a4c-4186-408d...	Goodman
1957-04-17	lettie.bennett@odeter...	Lettie	9baf9784-4141-4da2-...	Bennett
1944-03-24	mabel.leach@lisoheje....	Mabel	6eb50264-b0d0-480...	Leach
1980-05-05	jordan.miccinesi@duo...	Jordan	12d587f4-8a79-4f66-...	Miccinesi
1941-03-06	marie.parkes@nowuf...	Marie	6f65ba79-6088-45e2...	Parkes
1956-03-05	rose.gray@kagu.hr	Rose	3ad23545-3f43-4513-...	Gray
2006-12-15	garrett.stokes@fef.bg	Garrett	a0075d26-4477-4883...	Stokes
1927-12-11	barbara.matthieu@de...	Barbara	cbe6ebb9-96da-49a7...	Matthieu
1947-05-20	jean.rhodes@wehovu...	Jean	304a4ee0-8ad9-409...	Rhodes
1973-03-15	jack.romoli@zamum.b...	Jack	9470cc40-cc4a-4b43...	Romoli

The Vaadin way }>

Just Java

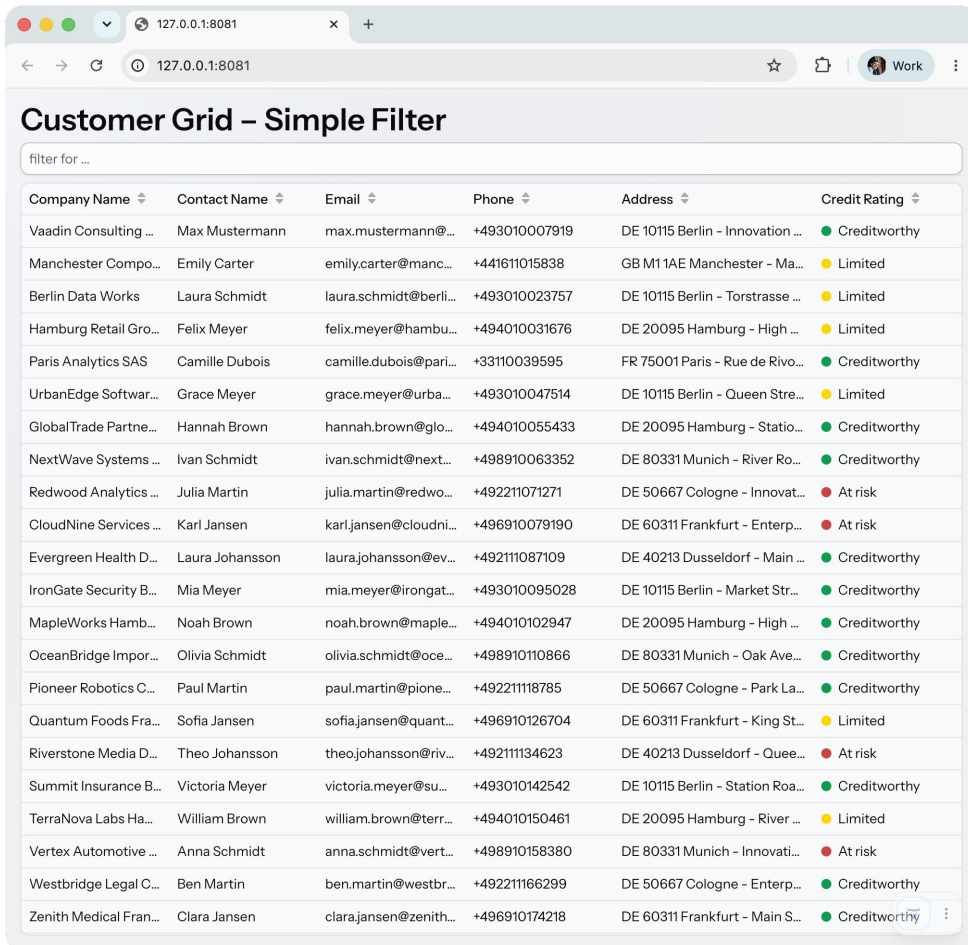
```
@Route(value = "")
public class HelloWorldView extends HorizontalLayout {

    public HelloWorldView(PersonRepository repository) {

        var grid = new Grid<>(Person.class);
        grid.setItems(repository.findAll());
        add(grid);
    }
}
```

Java only

Simple Filter



Company Name	Contact Name	Email	Phone	Address	Credit Rating
Vaadin Consulting ...	Max Mustermann	max.mustermann@...	+493010007919	DE 10115 Berlin - Innovation ...	Creditworthy
Manchester Compo...	Emily Carter	emily.carter@manc...	+441611015838	GB M1 1AE Manchester - Ma...	Limited
Berlin Data Works	Laura Schmidt	laura.schmidt@berli...	+493010023757	DE 10115 Berlin - Torstrasse ...	Limited
Hamburg Retail Gro...	Felix Meyer	felix.meyer@hambu...	+494010031676	DE 20095 Hamburg - High ...	Limited
Paris Analytics SAS	Camille Dubois	camille.dubois@pari...	+33110039595	FR 75001 Paris - Rue de Rivo...	Creditworthy
UrbanEdge Softwar...	Grace Meyer	grace.meyer@urba...	+493010047514	DE 10115 Berlin - Queen Stre...	Limited
GlobalTrade Partne...	Hannah Brown	hannah.brown@glo...	+494010055433	DE 20095 Hamburg - Statio...	Creditworthy
NextWave Systems ...	Ivan Schmidt	ivan.schmidt@next...	+498910063352	DE 80331 Munich - River Ro...	Creditworthy
Redwood Analytics ...	Julia Martin	julia.martin@redwo...	+492211071271	DE 50667 Cologne - Innovat...	At risk
CloudNine Services ...	Karl Jansen	karl.jansen@cloudni...	+496910079190	DE 60311 Frankfurt - Enterp...	At risk
Evergreen Health D...	Laura Johansson	laura.johansson@ev...	+492111087109	DE 40213 Dusseldorf - Main ...	Creditworthy
IronGate Security B...	Mia Meyer	mia.meyer@irongat...	+493010095028	DE 10115 Berlin - Market Str...	Creditworthy
MapleWorks Hamb...	Noah Brown	noah.brown@maple...	+494010102947	DE 20095 Hamburg - High ...	Creditworthy
OceanBridge Impor...	Olivia Schmidt	olivia.schmidt@oce...	+498910110866	DE 80331 Munich - Oak Ave...	Creditworthy
Pioneer Robotics C...	Paul Martin	paul.martin@pione...	+49221118785	DE 50667 Cologne - Park La...	Creditworthy
Quantum Foods Fra...	Sofia Jansen	sofia.jansen@quant...	+496910126704	DE 60311 Frankfurt - King St...	Limited
Riverstone Media D...	Theo Johansson	theo.johansson@riv...	+492111134623	DE 40213 Dusseldorf - Quee...	At risk
Summit Insurance B...	Victoria Meyer	victoria.meyer@su...	+493010142542	DE 10115 Berlin - Station Roa...	Creditworthy
TerraNova Labs Ha...	William Brown	william.brown@terr...	+494010150461	DE 20095 Hamburg - River ...	Limited
Vertex Automotive ...	Anna Schmidt	anna.schmidt@vert...	+498910158380	DE 80331 Munich - Innovati...	At risk
Westbridge Legal C...	Ben Martin	ben.martin@westbr...	+492211166299	DE 50667 Cologne - Enterp...	Creditworthy
Zenith Medical Fran...	Clara Jansen	clara.jansen@zenith...	+496910174218	DE 60311 Frankfurt - Main S...	Creditworthy

Java only

Simple Filter

```
var filterField = new TextField("", "filter for ...");
filterField.addValueChangeListener(this::onFilter);
add(filterField);

private void onFilter(ComponentValueChangeEvent<TextField, String> event) {
    var filteredCustomers = customerRepository.findAll().stream()
        .filter(customer -> customer.getCompanyName().toLowerCase()
            .contains(event.getValue().toLowerCase()) || ... ).toList();

    this.grid.setItems(filteredCustomers);
}
```

Vaadin + Spring Data JPA

Lazy Filter

Company Name	Contact Name	Email	Phone	Address	Credit Rating
Vaadin Consulting ...	Max Mustermann	max.mustermann...	+493010007919	DE 10115 Berlin - In...	Creditworthy
Manchester Comp...	Emily Carter	emily.carter@man...	+441611015838	GB M11AE Manch...	Limited
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Redwood Analytic...	Julia Martin	julia.martin@redw...	+492211071271	DE 50667 Cologne...	At risk
CloudNine Service...	Karl Jansen	karl.jansen@cloud...	+496910079190	DE 60311 Frankfurt...	At risk
Evergreen Health ...	Laura Johansson	laura.johansson@e...	+492111087109	DE 40213 Dusseld...	Creditworthy
IronGate Security ...	Mia Meyer	mia.meyer@ironga...	+493010095028	DE 10115 Berlin - M...	Creditworthy
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OceanBridge Impo...	Olivia Schmidt	olivia.schmidt@oc...	+498910110866	DE 80331 Munich - ...	Creditworthy
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Riverstone Media ...	Theo Johansson	theo.johansson@ri...	+492111134623	DE 40213 Dusseld...	At risk
Summit Insurance ...	Victoria Meyer	victoria.meyer@su...	+493010142542	DE 10115 Berlin - St...	Creditworthy
TerraNova Labs Ha...	William Brown	william.brown@ter...	+494010150461	DE 20095 Hambur...	Limited
Vertex Automotive	Anna Schmidt	anna.schmidt@ver...	+498910158380	DE 80331 Munich - ...	At risk

Lazy Filter

```
grid.setItems(  
    gridQuery -> customerRepository.findAll(buildCustomerSpecification(),  
        VaadinSpringDataHelpers.toSpringPageRequest(gridQuery)).stream(),  
    _ -> Math.toIntExact(customerRepository.count(buildCustomerSpecification())));  
  
private Specification<Customer> buildCustomerSpecification() {  
    return (root, query, criteriaBuilder) -> {  
        var predicates = new ArrayList<>();  
  
        if (filterCustomer.getCompanyName() != null && ...) {  
            predicates.add(criteriaBuilder.like(...
```


Filter with natural language

Multi-step: call tools, Pass Intermediate Results, Follow up

localhost:8083

localhost:8083

Customer Grid – AI Filter (simple)

filter for ...

Company Name	Contact Name	Email	Phone	Address	Credit Rating
Vaadin Consulting ...	Max Mustermann	max.mustermann...	+493010007919	DE 10115 Berlin - In...	● Creditworthy
Manchester Comp...	Emily Carter	emily.carter@man...	+441611015838	GB M11AE Manch...	● Limited
Berlin Data Works	Laura Schmidt	laura.schmidt@ber...	+493010023757	DE 10115 Berlin - T...	● Limited
Hamburg Retail Gr...	Felix Meyer	felix.meyer@hamb...	+494010031676	DE 20095 Hambur...	● Limited
Paris Analytics SAS	Camille Dubois	camille.dubois@pa...	+33110039595	FR 75001 Paris - R...	● Creditworthy
UrbanEdge Softwa...	Grace Meyer	grace.meyer@urba...	+493010047514	DE 10115 Berlin - Q...	● Limited
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Redwood Analytic...	Julia Martin	julia.martin@redw...	+492211071271	DE 50667 Cologne...	● At risk
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Evergreen Health ...	Laura Johansson	laura.johansson@e...	+492111087109	DE 40213 Dusseld...	● Creditworthy
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Riverstone Media ...	Theo Johansson	theo.johansson@ri...	+492111134623	DE 40213 Dusseld...	● At risk
Summit Insurance ...	Victoria Meyer	victoria.meyer@su...	+493010142542	DE 10115 Berlin - St...	● Creditworthy
TerraNova Labs Ha...	William Brown	william.brown@ter...	+494010150461	DE 20095 Hambur...	● Limited
Vertex Automotive...	Anna Schmidt	anna.schmidt@ver...	+498910158380	DE 80331 Munich - ...	● At risk

Prompt

```
chatClient.prompt()  
    .system("""  
        You are a helpful assistant that helps users find customers ...  
        """)  
    .user(event.getValue())  
    .tools(this)  
    .stream()  
    .content()  
    .subscribe(token -> {}); //do something in the UI
```

Tool Call

```
/** DON'T USE THIS! IT IS AN ANTI-PATTERN **/  
@Tool(description = "Returns all customers with company name, ...")  
List<Customer> customerList() {  
    return customerRepository.findAll();  
}
```

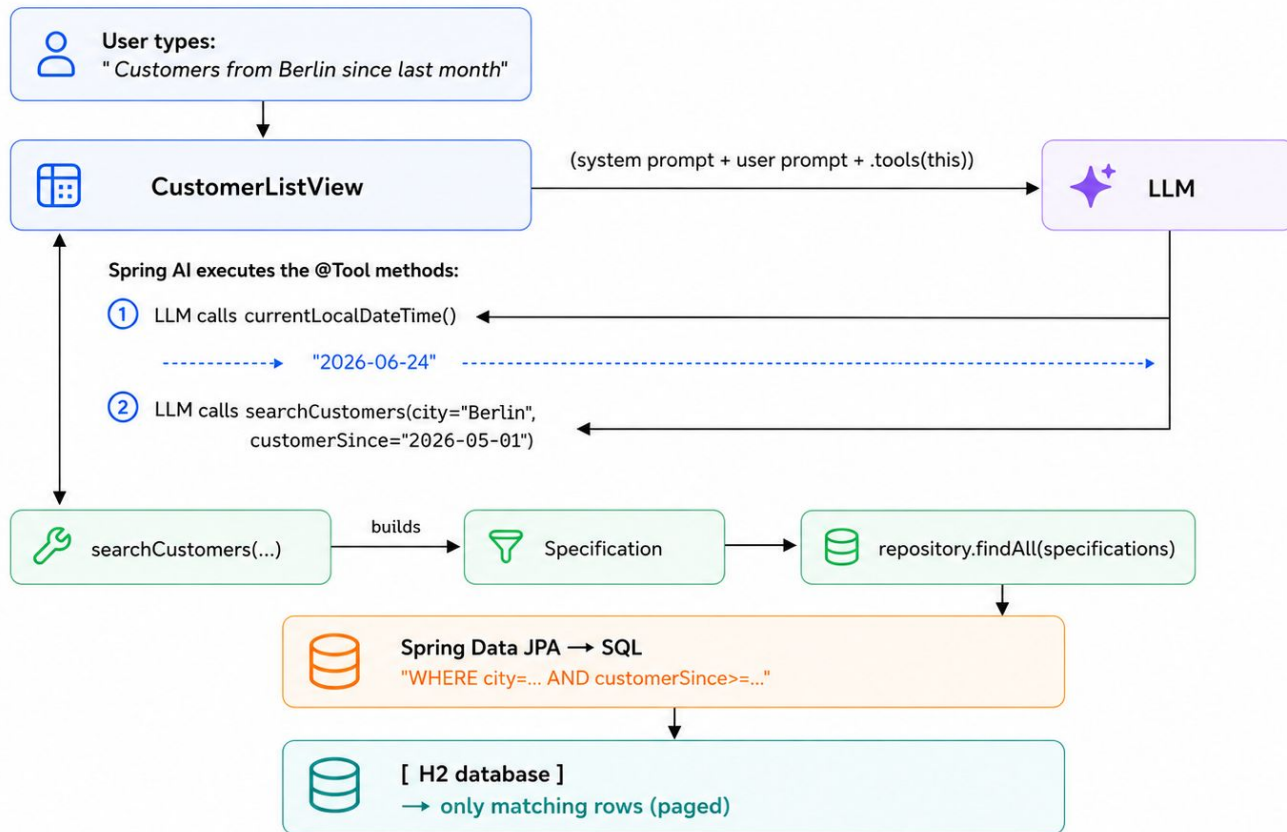
“Filter a company named Vaadin” → `searchCustomers(“Vaadin”, ...)`

Tool Call

```
@Tool(description = "Search and filter the customer grid. Returns nothing; it ... ")
void searchCustomers(@ToolParam(description = "part name of ...") String companyName, ...) {

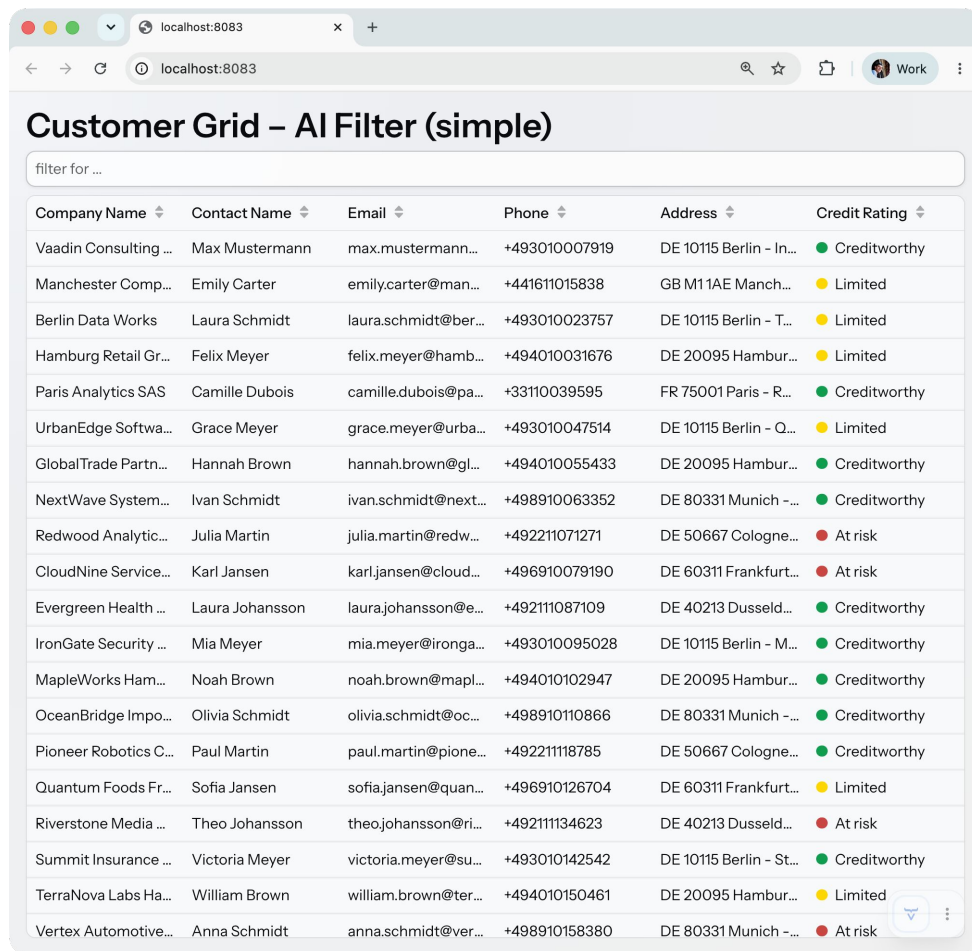
    getUI().ifPresent(ui -> ui.access(() -> {
        var customerSpecification = buildCustomerSpecification(companyName, ...);
        grid.setItems(
            query -> customerRepository.findAll(customerSpecification,
                VaadinSpringDataHelpers.toSpringPageRequest(query)).stream(),
            _ -> Math.toIntExact(customerRepository.count(customerSpecification))
        );
    }));
}
```

Agentic Filtering



The LLM returns structured JSON

One call, One Filter Object



The screenshot shows a web browser window with the address bar displaying 'localhost:8083'. The page title is 'Customer Grid - AI Filter (simple)'. Below the title is a search bar labeled 'filter for ...'. The main content is a table with 6 columns: Company Name, Contact Name, Email, Phone, Address, and Credit Rating. The table contains 20 rows of customer data. The Credit Rating column uses colored dots to indicate the status: green for 'Creditworthy', yellow for 'Limited', and red for 'At risk'.

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Riverstone Media ...	Theo Johansson	theo.johansson@ri...	+492111134623	DE 40213 Dusseld...	● At risk
Summit Insurance ...	Victoria Meyer	victoria.meyer@su...	+493010142542	DE 10115 Berlin - St...	● Creditworthy
TerraNova Labs Ha...	William Brown	william.brown@ter...	+494010150461	DE 20095 Hambur...	● Limited
Vertex Automotive...	Anna Schmidt	anna.schmidt@ver...	+498910158380	DE 80331 Munich - ...	● At risk

local LLMs to produce reliably as structured output

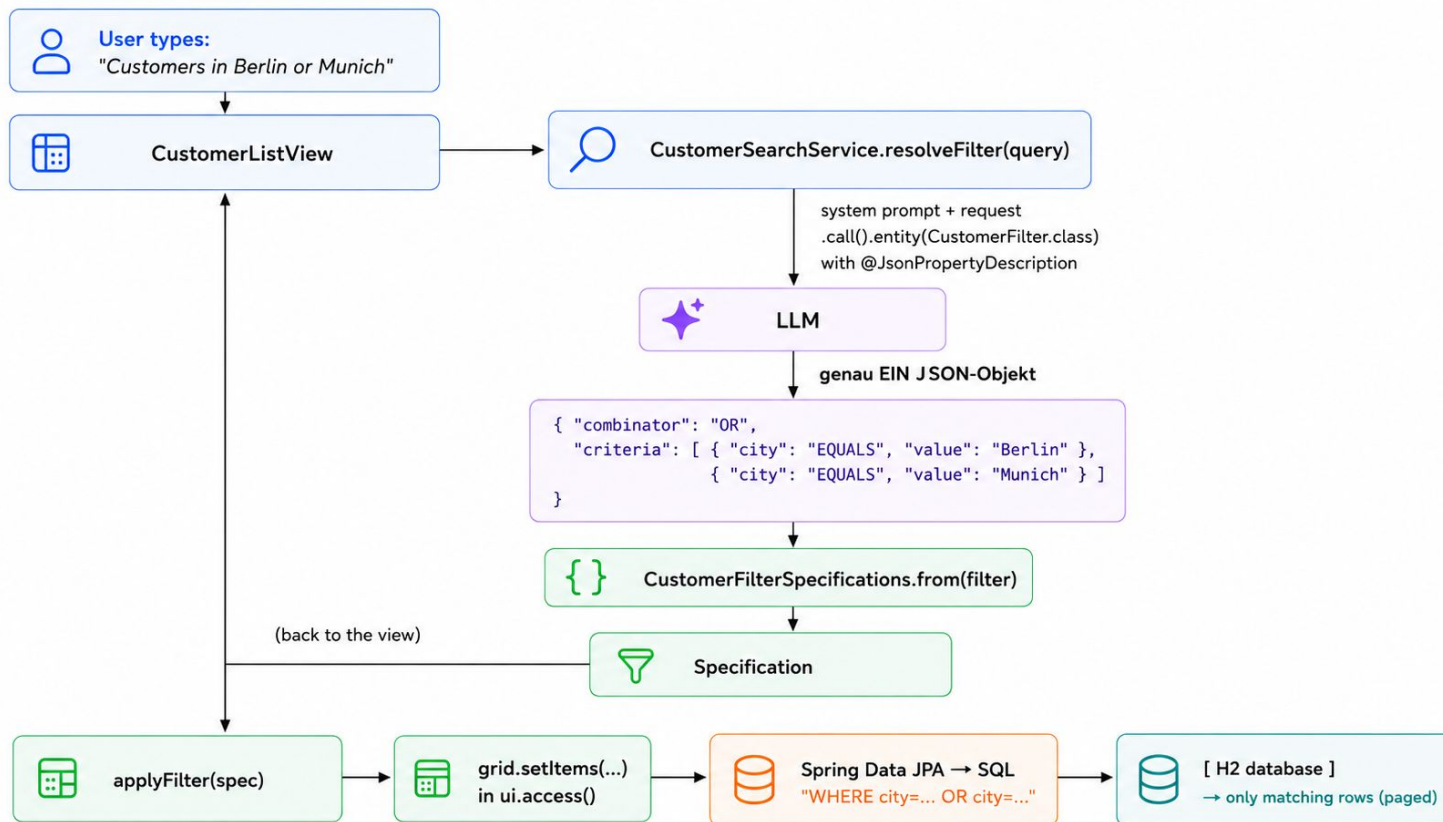
JSON Description

```
@JsonClassDescription("A flat customer filter: a list of ...")  
public record CustomerFilter(  
    @JsonPropertyDescription("the conditions to ...") List<FilterCriterion> criteria) {}
```

Prompt

```
chatClient.prompt()  
    .advisors(SimpleLoggerAdvisor.builder().build())  
    .system(systemPrompt(LocalDate.now()))  
    .user(naturalLanguageQuery)  
    // Low temperature → deterministic structure, fewer formatting slips.  
    .options(ChatOptions.builder())  
    .call()  
    .entity(CustomerFilter.class);
```


Structured Output Filtering



Finding a **capable** local Model



A benchmark of 11 local models for accuracy, latency, and TTFT

More parameters, less performance

Model	Accuracy	Median latency	TTFT	Tokens/s	Wall-clock
qwen3:8b	22/22	1008 ms	271 ms	31.6	129.3 s
qwen3.5:4b-mlx	21/22	825 ms	69 ms	44.9	133.8 s
qwen3.5:9b-mlx	22/22	2011 ms	110 ms	27.1	276.3 s
gemma4:e4b-mlx	18/22	943 ms	108 ms	42.6	126.9 s
gemma4:12b-mlx	22/22	1708 ms	468 ms	20.1	295.8 s
gemma4:26b-mlx	21/22	843 ms	276 ms	39.5	139.2 s
llama3.1:8b	22/22	810 ms	277 ms	34.5	119.2 s
ministral-3:8b	20/22	1231 ms	337 ms	30.9	159.4 s
phi4-mini:3.8b	18/22	850 ms	238 ms	55.9	126.4 s
deepseek-r1:8b	5/22	16784 ms	12090 ms	30.8	2130.7 s
openai/gpt-5.4-mini	22/22 (1 err)	819 ms	512 ms	39.0	171.7 s

* executed in 5 runs on a MacBook Pro M2 pro with 32gb RAM

Testing the Application

**Two worlds:
deterministic code vs.
non-deterministic LLM**

FAST, no LLM, no Docker

1. Deterministic Layer

- `@DataJpaTest` on seeded H2
- **Tests** `CustomerFilter` → JPA Specification
- Pins the rules: same field
→ OR, different fields
→ AND
- Runs on every `./mvnw test`.

✓	CustomerFilterSpecificationsTest (dev.de 469 ms)	
✓	rangeOnSameNumericFieldsAnd()	315 ms
✓	creditRatingCombinesWithCityAsAnd()	14 ms
✓	twoCreditRatingsAreOr()	13 ms
✓	emptyFilterMatchesAll()	105 ms
✓	notEqualsExcludesCity()	9 ms
✓	sameFieldsOr_differentFieldsAnd()	8 ms
✓	singleCity()	5 ms

2. AI Layer

- Filter string → real Ollama
- Assert on structure, not wording:
`hasCriterion(field, operator, value)`, extras ignored
- Relative dates → check the *shape* only
(`lastOrderDate ≥ ...`)
- Two backends:
 - native Ollama
 - Testcontainer

✓	CustomerSearchServiceLocalOllama	51 sec 609 ms
✓	phoneNumberContains()	10 sec 956 ms
✓	contactNameStartsWith()	2 sec 916 ms
✓	citiesAndRevenue_keepsEveryCo	4 sec 612 ms
✓	singleFalseCity()	2 sec 736 ms
✓	contactNameEndsWith()	2 sec 950 ms
✓	multipleCities()	3 sec 576 ms
✓	contactNameAndCity()	3 sec 734 ms
✓	singleCity()	2 sec 732 ms
✓	contactNameContains()	2 sec 911 ms
✓	orderedInTheLastWeek()	14 sec 486 ms

To be defined, discussed & developed

Open Points

- More test cases (e.g. check other languages, more complex queries)
- Run Filter Tests for AI in Benchmark Mode
- Add more parameters to Benchmark
- Cascade more AND and OR cases in the Filter for complex queries
- Integrating and testing Guardrails
- Create a development process for Ollama tests and claude code (run test cases on Ollama and claude code improve context information of tools, etc.)
- Add other use cases in the prompt
 - "Highlight all items with good credit score green"*
 - "Sort ascending by City and Revenue"*
 - "Show only names and address of customers"*
 - "Show top 10 customers by revenue"*

Try it out!

Sources available on GH



Slides
Demo Project

seb@vaadin.com

[http://github.com/
SebastianKuehnau/
vaadin-grid-ai-filter](http://github.com/SebastianKuehnau/vaadin-grid-ai-filter)

Meet us at
Vaadin Create 2026
in Barcelona
Oct 27-28, 2026

Discount Code:
JUGSUMMER26

[vaadin.com/
vaadin-create-2026](https://vaadin.com/vaadin-create-2026) vaadin}>

Thank you

**We would
like to talk.
Contact us!**

seb@vaadin.com

[linkedin.com/in/sebastiankuehnau](https://www.linkedin.com/in/sebastiankuehnau)

github.com/SebastianKuehnau

vaadin.com/forum